

```

/* The Bogo Sort "Algorithm":
 1) Scramble the contents.
 2) Check to see if the contents are sorted.
 3) If not, goto step 1.
*/

public class BogoSort
{
    void bogo(int[] arr) {
        //Keep a track of the number of shuffles
        int shuffle=1;
        while(!isSorted(arr)) {
            shuffle(arr);
            ++shuffle;
        }

        //Boast
        System.out.println("This took "+shuffle+" shuffles.");
    }

    void shuffle(int[] arr) {
        //Standard Fisher-Yates shuffle algorithm
        int i=arr.length-1;
        while(i>0)
            swap(arr,i--, (int)(Math.random()*i));
    }

    void swap(int[] arr,int i,int j) {
        int temp=arr[i];
        arr[i]=arr[j];
        arr[j]=temp;
    }

    boolean isSorted(int[] arr) {
        for(int i=1;i<arr.length;i++)
            if(arr[i]<arr[i-1])
                return false;
        return true;
    }

    void display1D(int[] arr) {
        for(int i=0;i<arr.length;i++)
            System.out.print(arr[i]+" ");
        System.out.println();
    }

    public static void main(String[] args) {
        //Enter array to be sorted here
        int[] arr={4,5,6,0,7,8,9,1,2,3};

        BogoSort now=new BogoSort();
        System.out.print("Unsorted: ");
        now.display1D(arr);

        now.bogo(arr);

        System.out.print("Sorted: ");
        now.display1D(arr);
    }
}

```